

Arduino Keypad and LCD Interface Project

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July 15, 2026

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1 Project Overview

This project demonstrates interfacing a 4x4 matrix keypad with an Arduino Uno and a 16x2 LCD display.

The Arduino reads the pressed key from the keypad and displays the detected character on the LCD screen.

2 Required Components

- Arduino Uno
- 4x4 Matrix Keypad
- 16x2 LCD Display
- Breadboard
- Jumper Wires

3 Circuit Connections

3.1 LCD Connections

LCD Pin	Arduino Pin
RS	12
EN	11
D4	5
D5	4
D6	3
D7	2

3.2 Keypad Connections

Keypad Pin	Arduino Pin
Row 1	A0
Row 2	A1
Row 3	A2
Row 4	A3
Column 1	6
Column 2	7
Column 3	8
Column 4	9

4 Working Principle

The keypad works based on row and column scanning.

When a button is pressed, Arduino detects the corresponding row and column and displays the pressed key on the LCD.

5 Arduino Source Code

```
1
2      #include <Keypad.h>
3      #include <LiquidCrystal.h>
4
5      LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
6
7
8      const byte ROWS = 4;
9      const byte COLS = 4;
10
11
12      char keys[ROWS][COLS] = {
13          {'1','2','3','A'},
14          {'4','5','6','B'},
15          {'7','8','9','C'},
16          {'*','0','#','D'}
17      };
18
19
20      byte rowPins[ROWS] = {
21          A0, A1, A2, A3
22      };
23
24
25      byte colPins[COLS] = {
26          6, 7, 8, 9
27      };
28
29
30      Keypad keypad = Keypad(
31          makeKeymap(keys),
32          rowPins,
33          colPins,
34          ROWS,
35          COLS
36      );
37
38
39      void setup()
40      {
41          lcd.begin(16,2);
42          lcd.print("Press Key");
43      }
44
45
46      void loop()
47      {
48
49          char key = keypad.getKey();
```

```

50
51
52         if (key)
53         {
54
55             lcd.clear();
56
57             lcd.setCursor(0,0);
58             lcd.print("You Pressed:");
59
60             lcd.setCursor(0,1);
61             lcd.print(key);
62
63         }
64
65     }

```

6 Expected Output

After starting the system, LCD displays:

Press Key

The tested keys are:

- Key 4
- Key 6
- Key A
- Key B

Example outputs:

You Pressed:

4

You Pressed:

6

You Pressed:

A

You Pressed:

B

7 Project Images

Image 1

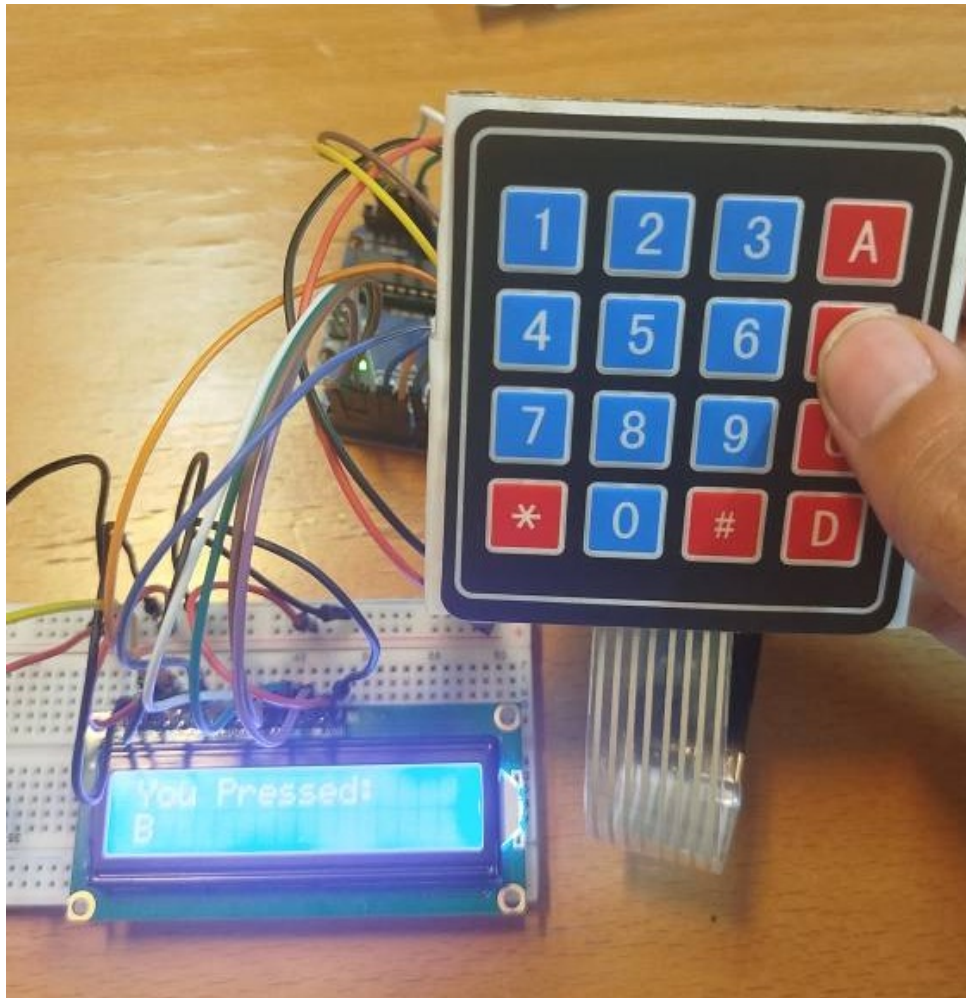


Image 2

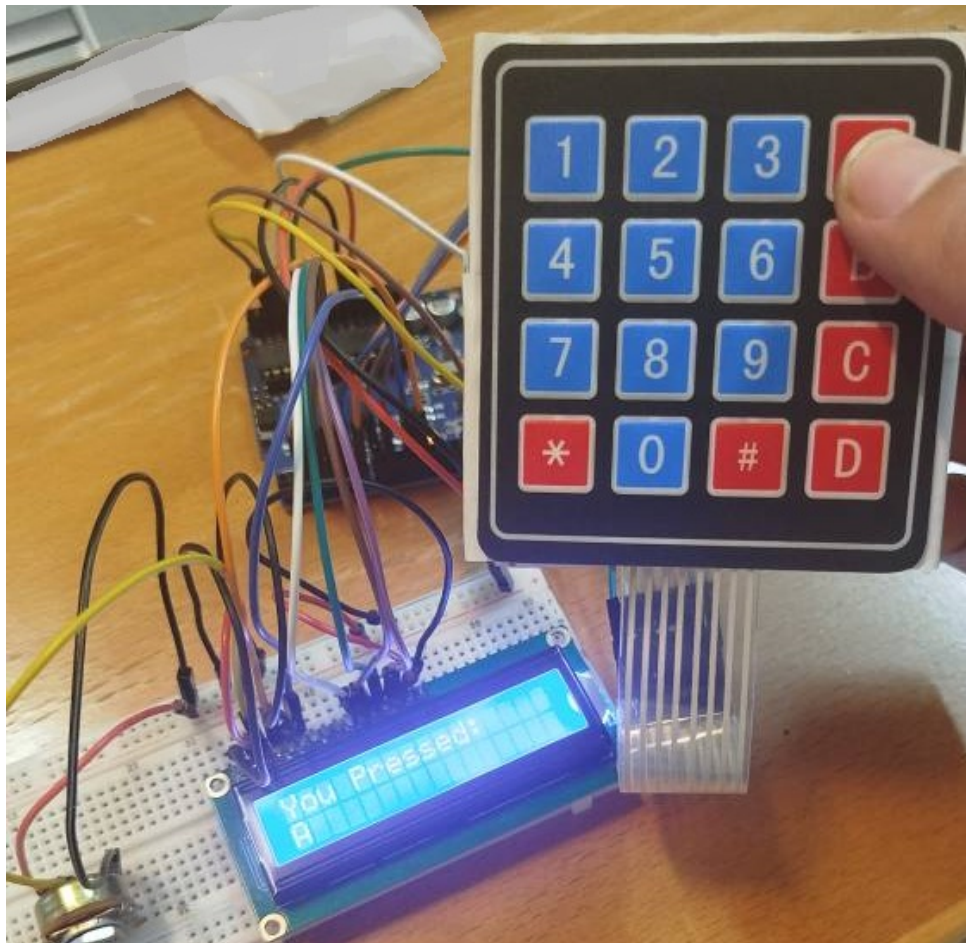


Image 3

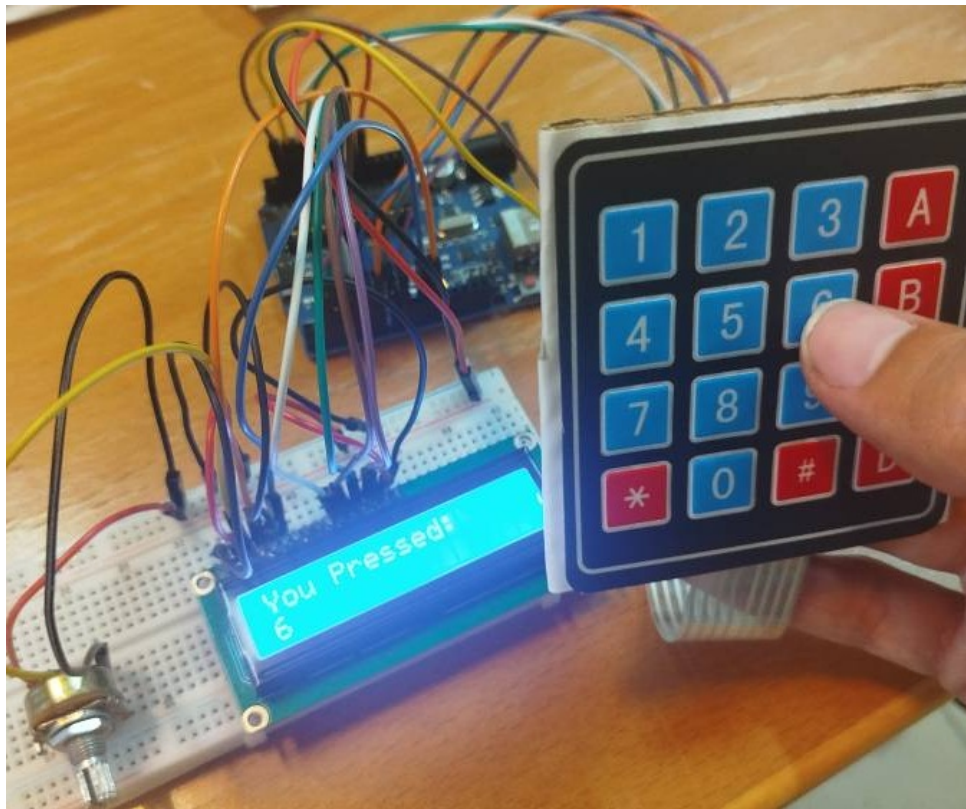
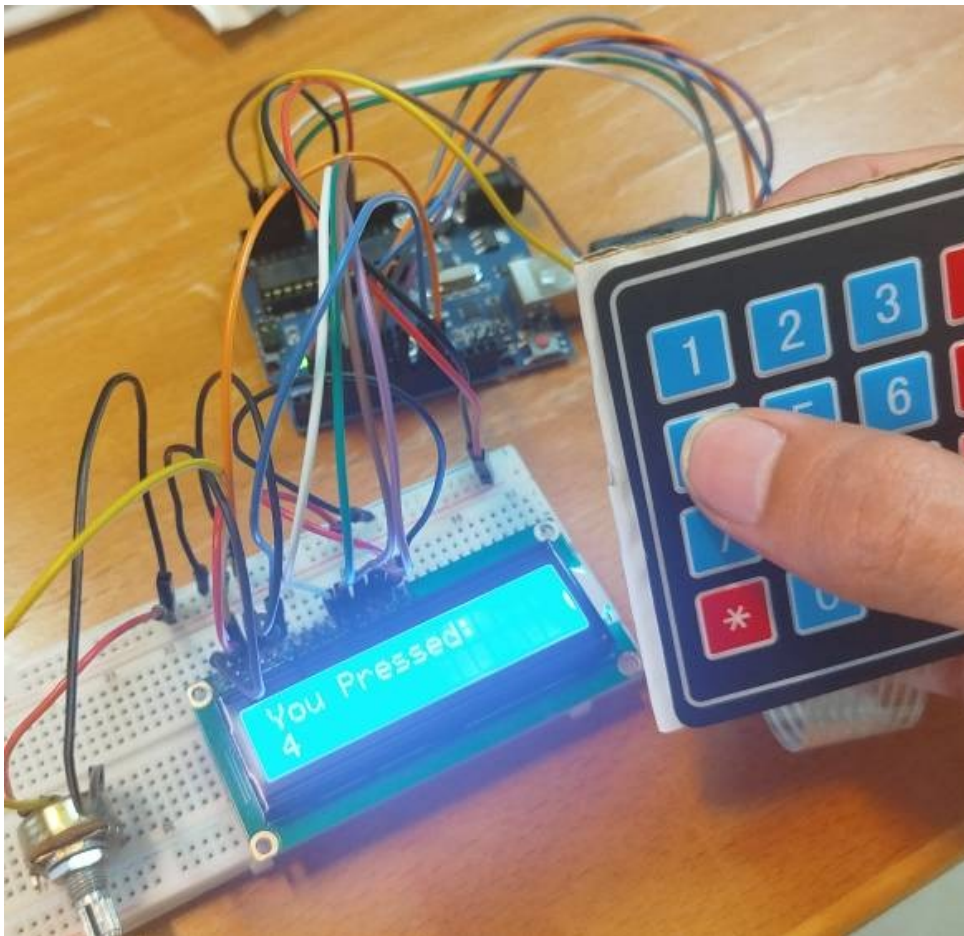


Image 4



8 Applications

- Password Security Systems
- Electronic Door Locks
- ATM Interfaces
- Calculator Systems
- Embedded User Interfaces

9 Conclusion

This project demonstrates communication between Arduino Uno, a 4x4 matrix keypad, and a 16x2 LCD display.

The system introduces keypad scanning, digital input detection, and LCD communication in embedded systems.